

Reducing CO₂ Levels Using Genetic Algorithms: A Re-evaluation

Cristian Lincu

Abstract

This project provides a solution framework for minimizing CO₂ emission levels in a national power grid through machine learning methods. The data is collected via Energinet’s public API, which provides real-time data about Denmark’s power system. The framework retains the structure of the original formulation — a regression model inferring CO₂ levels, gradient-boosted regressors forecasting demand and renewable production, and a genetic ensemble searching for the emission-minimizing distribution — while correcting a set of modelling errors that made the original results unreliable. The central correction is that the optimizer was previously able to exploit its own surrogate: an unconstrained regression tree, fitted to confounded data, reported that emissions *fall* when thermal generation rises, and the genetic heuristic duly proposed ramping thermal plants. This work constrains the emissions model to be physically monotone, evaluates every component chronologically rather than on shuffled splits, replaces scale-invariant coherence with explicit ramp limits, and measures the optimizer against the counterfactual in which dispatch is held fixed. The resulting system proposes feasible four-hour dispatch trajectories that reduce CO₂ intensity by approximately 55% relative to that counterfactual.

1 Introduction

The data is collected via Energinet’s public API¹ which provides real-time data about Denmark’s power system. The dataset prepared for this research consists of one year of records aggregated onto the 15-minute grid used for imbalance settlement in Denmark, giving 34,600 observations between September 2025 and September 2026.

The main elements of this project are: a regression model for inferring CO₂ levels, XGBoost regressors for predicting demand and renewable energy production, and genetic algorithms for finding the optimal energy distribution with respect to CO₂ emission minimization. The goal is to infer the optimal resource combinations that minimize CO₂ emissions, depending on future demand and renewables production.

This paper is a re-evaluation of an earlier version of the same framework. The earlier version reported potential emission reductions of 50–75%. Those figures did not survive scrutiny, for reasons set out in Section 5. The methodology below preserves as much of the original design as the physics allows, and states plainly where it does not.

2 Data and the energy balance

The national energy balance is the identity

$$D = \underbrace{P_{\geq 100} + P_{< 100}}_{\text{generation}} + R + \sum_{i \in \mathcal{B}} x_i, \quad (1)$$

where R denotes renewable production (solar and wind), $P_{\geq 100}$ and $P_{< 100}$ denote energy produced by power plants with installed capacity greater or equal to 100 MW and less than 100 MW respectively, and $\mathcal{B}$ is the set of *cross-border* interconnectors: DK1-DE, DK1-NL, DK1-GB,

¹<https://www.energidataservice.dk/tso-electricity/PowerSystemRightNow>

DK1-NO, DK1-SE, DK2-DE, DK2-SE and Bornholm-SE. Positive exchange values are imports, negative ones are exports.

The DK1-DK2 link is deliberately absent from $\mathcal{B}$. It is the Great Belt Power Link, which joins the two Danish bidding zones; power flowing across it is internal and cannot change national supply. Energinet excludes it from its own `Exchange_Sum` field for precisely this reason, which is verifiable directly from the API: on an arbitrary record the eight cross-border flows sum to 3481.49 MW and `Exchange_Sum` reports 3481.49 MW, with the DK1-DK2 value of -371.57 MW sitting outside the total. The original formulation included DK1-DK2 in the balance, displacing demand by up to 600 MW against a tolerance of 1% of demand. It remains a decision variable, because it carries zonal information the emissions model uses, but it no longer contributes to Equation (1).

3 Methodology

3.1 I. Inferring CO₂ levels

A regression model infers CO₂ levels taking into account 12 features: renewable energy production (solar and wind), energy produced by power plants with installed capacity greater or equal to 100 MW, energy produced by power plants with installed capacity less than 100 MW, and the nine energy exchanges between Denmark and other countries or areas. The hyperparameters are tuned using randomized search, and the model is evaluated through time-series cross-validation.

Two changes to this component are essential rather than cosmetic.

The target is total emissions, not intensity. Energinet publishes CO₂ as an intensity in g/kWh. Modelling intensity directly is what allowed the original surrogate to be exploited, because intensity is confounded: in Denmark, hours with high output from large thermal plants are also hours with high wind, so an unconstrained model learns that thermal generation predicts *low* emissions. The model here predicts the total emission rate

$$\Theta = \frac{\text{CO}_2 \text{ intensity} \times D}{1000} \quad [\text{t/h}], \quad (2)$$

from which intensity is recovered as $1000 \Theta / D$. Under consumption-based accounting every supply source carries a non-negative emission factor, so total emissions are monotonically non-decreasing in every source. That is imposed as a hard constraint on all eleven controllable features. Renewables are left unconstrained, since they are not a decision variable.

The estimator is an ensemble, and it reports its own reliability. A single constrained regression tree lacks the capacity to fit the data without leaning on the confound, so the deployed estimator is a gradient-boosted ensemble of regression trees carrying the same monotonicity. A companion depth-capped tree partitions the feature space and supplies, for any candidate, the number of training observations $n(\mathbf{x})$ behind its leaf and the dispersion $\sigma(\mathbf{x})$ of the target within it. A Mahalanobis distance $d^2(\mathbf{x})$ to the training distribution measures how far a candidate sits from anything ever observed. Both quantities enter the fitness function, so the optimizer is penalized for leaving the region where the model has evidence.

3.2 II. Forecasting demand and renewable production

The XGBoost regressors provide multi-step forecasting for energy demand and renewable production over the next 16 time points, that is four hours ahead at 15-minute resolution. These

forecasts are important because the optimal power distribution has to match demand continuously; a persistent mismatch between supply and demand is what a system operator resolves through reserves, and the schedule the optimizer proposes must not create one. The distributable energy whose allocation is optimized is the difference between forecasted demand and forecasted renewable energy, $D_{XG} - R_{XG}$.

Renewable production is no longer extrapolated from its own recent output. Wind four hours ahead is not recoverable from the last few observations, and attempting it was the weakest part of the original system. Instead the model consumes Energinet’s own physical forecast and learns a correction on top of it. The **Forecast5Hour** vintage is issued approximately five hours before its target and is therefore genuinely available at decision time across the whole horizon, which keeps the evaluation free of look-ahead.

Every model is trained on a chronological split and reported against a persistence baseline, and the renewables model additionally against the raw Energinet forecast it post-processes.

3.3 III. Genetic optimization of a dispatch trajectory

For the forecasted horizon, the optimal combination of sources is inferred using genetic algorithms. In order to escape eventual local optima and increase the method’s success, this is done by employing a genetic ensemble with 16 optimizers. Populations of solutions are initialized stochastically and transformed by evolutionary processes to arrive at the “fittest” distribution. The restrictions are the ranges of power plant production and the interconnector limits for exchanges, inferred from the past year’s data and capped by technical transfer capacity. Ramp limits are taken from the 99th percentile of realised 15-minute changes over the same year, which describes what the fleet can actually do better than any fixed assumption. At 1 MW resolution a single time step admits approximately 8×10^{34} combinations.

A chromosome is a trajectory. In the original formulation each forecasted time point was optimized independently, and coherence between consecutive solutions was encouraged by a cosine-similarity term. Cosine similarity is scale invariant — two distributions differing by a factor of ten have similarity 1 — so it could not constrain magnitudes, and in the deployed system it did not: consecutive five-minute solutions differed by more than 1,100 MW on a single interconnector. Here a chromosome is the whole trajectory. Writing $d = 11$ for the number of genes per time step and $T = 16$ for the horizon, each individual is a matrix $X^{(j)} \in \mathbb{R}^{T \times d}$ whose rows $x_t^{(j)}$ are the distributions at each step, and the first generation is

$$G = \begin{bmatrix} - & \text{vec}(X^{(1)}) & - \\ - & \text{vec}(X^{(2)}) & - \\ & \vdots & \\ - & \text{vec}(X^{(m)}) & - \end{bmatrix} \in \mathbb{R}^{m \times Td}, \quad (3)$$

where m is the size of the population. Coherence is then imposed directly, as a ramp constraint between consecutive rows, rather than hoped for through a similarity term.

The fitness function. Each chromosome is evaluated using the regression ensemble to infer the CO₂ emission level implied by the solution it represents, while solutions that do not balance the power grid are penalized by an augmentation that significantly reduces their coupling chances. Each optimal trajectory minimizes

$$X^* = \underset{X^{(j)}}{\operatorname{argmin}} \frac{1}{T} \sum_{t=1}^T \left[\xi_b \mathcal{P}_b(x_t) + \xi_r \mathcal{P}_r(x_t, x_{t-1}) + \xi_s \mathcal{P}_s(x_t) + \zeta \Theta(x_t) + \gamma \sigma(x_t) \right], \quad (4)$$

whose terms are, in order,

- the **grid imbalance penalty**

$$\mathcal{P}_b(x_t) = \mathbf{I}\left(\left|D_{XG,t} - R_{XG,t} - \sum_{i \in \mathcal{B}} x_{t,i}\right| > \tau D_{XG,t}\right) \left(1 + \frac{|D_{XG,t} - R_{XG,t} - \sum_{i \in \mathcal{B}} x_{t,i}|}{\tau D_{XG,t}}\right), \quad (5)$$

where $\mathbf{I}$ is the indicator function, D_{XG} is the demand forecast and R_{XG} the renewable forecast provided by the XGBoost regressors, and $\tau = 0.01$. The indicator alone, as used originally, produces a flat plateau that a heuristic cannot descend; scaling it by the relative size of the violation gives the search a gradient towards feasibility while preserving the hard threshold;

- the **ramp penalty** $\mathcal{P}_r(x_t, x_{t-1}) = \sum_i \max(0, |x_{t,i} - x_{t-1,i}| - \rho_i) / \rho_i$, with ρ_i the per-resource ramp limit and x_0 the latest observed distribution. This term replaces the cosine similarity of the original formulation and performs the role that term was intended to perform;
- the **support penalty** $\mathcal{P}_s(x_t) = \mathbf{I}(n(x_t) < n_{\min}) + \max(0, d^2(x_t)/d_q^2 - 1)$, which confines the search to the region where the emissions model has evidence, with $n_{\min} = 20$ and d_q^2 the 99th percentile of the training Mahalanobis distances;
- the **emissions level** Θ inferred by the regression ensemble, augmented by ζ ; and
- the **model dispersion** σ , augmented by γ , which makes the optimizer prefer solutions the model is confident about.

Feasibility by construction. Every candidate is passed through a repair operator before evaluation. Walking the horizon forward, the admissible box at each step is the intersection of the technical limits with the ramp window around the previous step; the residual energy imbalance is then distributed across the balance-carrying resources in proportion to their remaining headroom. This turns feasibility from something the search must rediscover into a property of the representation, so the evolutionary effort is spent on emissions instead.

Implementation. The ensemble of independent optimizers is evaluated as a single batched array, an island model with periodic migration of the best individual between islands. Emissions, dispersion, leaf support and novelty all come out of one traversal per generation. With 16 islands, populations of 150 and 200 generations — 4.8×10^5 evaluations — a full solve takes approximately 57 seconds, which fits comfortably inside a scheduled job refreshing every 30 minutes.

4 Results

4.1 The emissions model

Table 1 reports the emissions model against the original formulation on identical data. Two comparisons matter. The original tree evaluated with a shuffled split achieves 5.77 g/kWh, but a shuffled split on a 15-minute series places near-duplicate neighbours on both sides of the partition; evaluated chronologically, the same model achieves 13.01 g/kWh. The constrained ensemble achieves 11.05 g/kWh — better than the original managed under honest evaluation, while additionally being physically valid.

Figure 1 shows why the constraint matters more than the accuracy. Holding every other resource fixed and sweeping large-plant output from 0 to 1500 MW, the original model reports total emissions falling to a minimum at 300 MW before rising. A genetic algorithm minimizing

Model	Evaluation	Intensity MAE (g/kWh)	Monotone
Fully grown tree, intensity	shuffled split	5.77	no
Fully grown tree, intensity	chronological	13.01	no
Monotone ensemble, total emissions	chronological	11.05	yes

Table 1: CO₂ model accuracy. The shuffled-split figure is reported because it is what the original framework published; it is inflated by leakage between adjacent observations.

that surface will ramp thermal generation, which is exactly what the deployed system did. The constrained model is monotone by construction and its minimum is at zero.

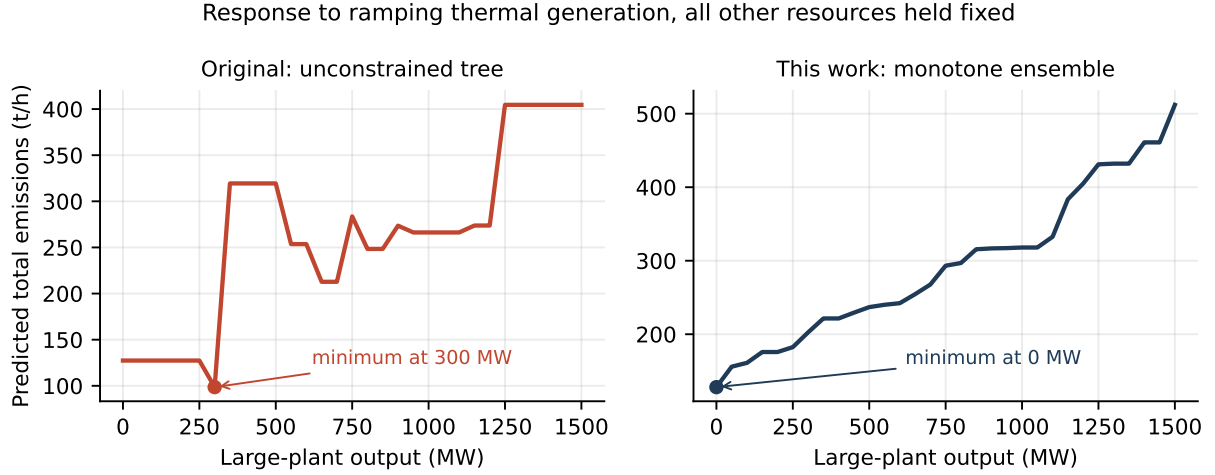


Figure 1: Response of each emissions model to ramping thermal generation with all other resources held fixed. The original model rewards the optimizer for starting thermal plants.

4.2 The forecasts

Figure 2 reports forecast error against horizon on a chronological hold-out. Demand RMSE grows from 88.9 MW at 15 minutes to 374.5 MW at four hours, against a persistence baseline of 98.3 MW and 815.8 MW. Renewables RMSE grows from 135.9 MW to 468.6 MW, against 151.7 MW and 1394.0 MW for persistence and a flat 699.5 MW for the raw Energinet forecast; post-processing that forecast with recent observations improves on it at every horizon.

4.3 The optimizer

An evolutionary heuristic is only worth its cost if it beats random search under the same evaluation budget. Figure 3 shows that it does: at 4.8×10^5 evaluations and matched wall-clock time, the genetic ensemble reaches a mean intensity of 60.8 g/kWh against 87.7 g/kWh for random search, and does so while keeping every step inside the region where the emissions model has support, which random search does not.

4.4 How the reduction is measured

The original framework compared optimized future emissions against the *present* measured value. That comparison credits the optimizer with whatever the weather was going to do anyway: if wind is rising over the horizon, intensity falls without any intervention at all.

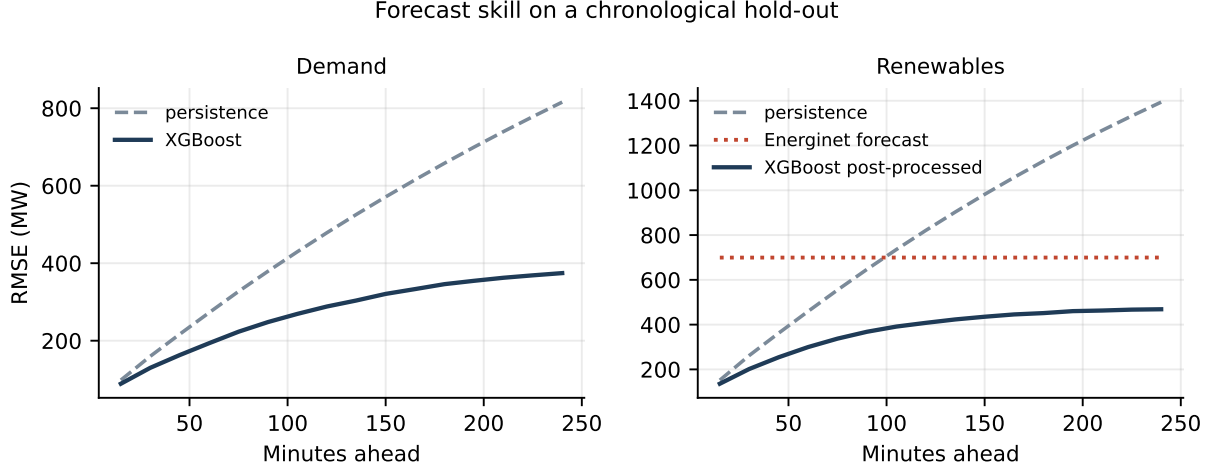


Figure 2: Forecast skill against horizon, with baselines.

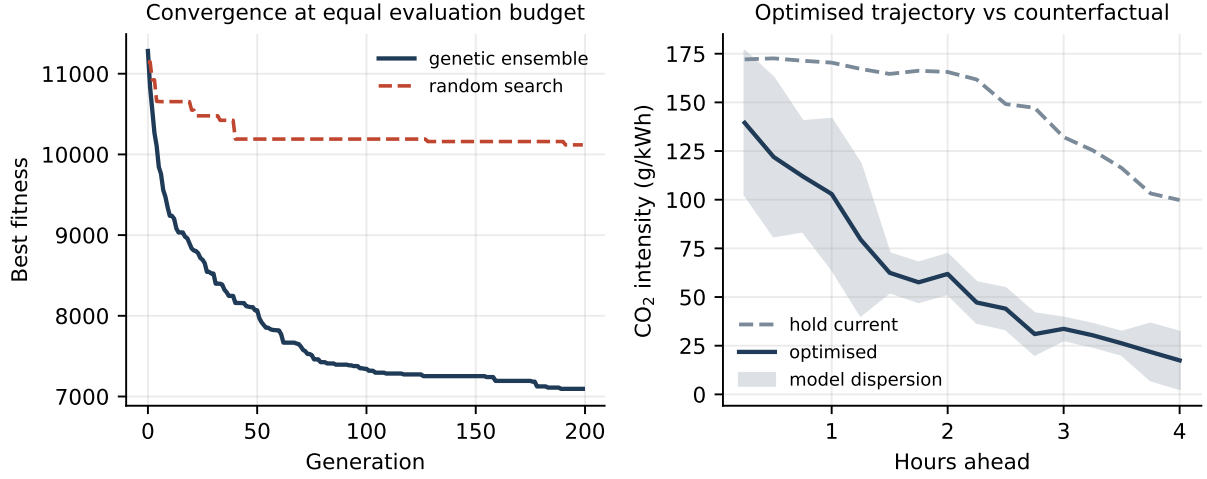


Figure 3: Left: convergence against random search at equal evaluation budget. Right: the optimized intensity trajectory against the hold-current counterfactual, with the emissions model's own dispersion shaded.

The reduction reported here is measured against a hold-current counterfactual, in which demand and renewables follow their forecasts but the controllable resources stay where they are. On the representative solve shown in Figure 4, the present measured intensity was 204.1 g/kWh, the counterfactual averaged 140.0 g/kWh over the horizon, and the optimized trajectory averaged 63.8 g/kWh. The honest reduction attributable to re-dispatch is therefore 54.4%, not the 68.7% that comparison against the present value would have produced.

The trajectory itself is physically legible, which the original solutions were not. Every step satisfies the energy balance to within a fraction of a megawatt and every change respects its ramp limit; the maximum balance error over the horizon is 2.4×10^{-4} MW.

5 What changed relative to the original formulation

Two further errors are worth recording because they were silent. The `PowerSystemRightNow` schema changed after the original models were trained: the `ImbalanceDK1`, `ImbalanceDK2` and `mFRRActivated` fields no longer exist. The original notebooks selected features by position, so the

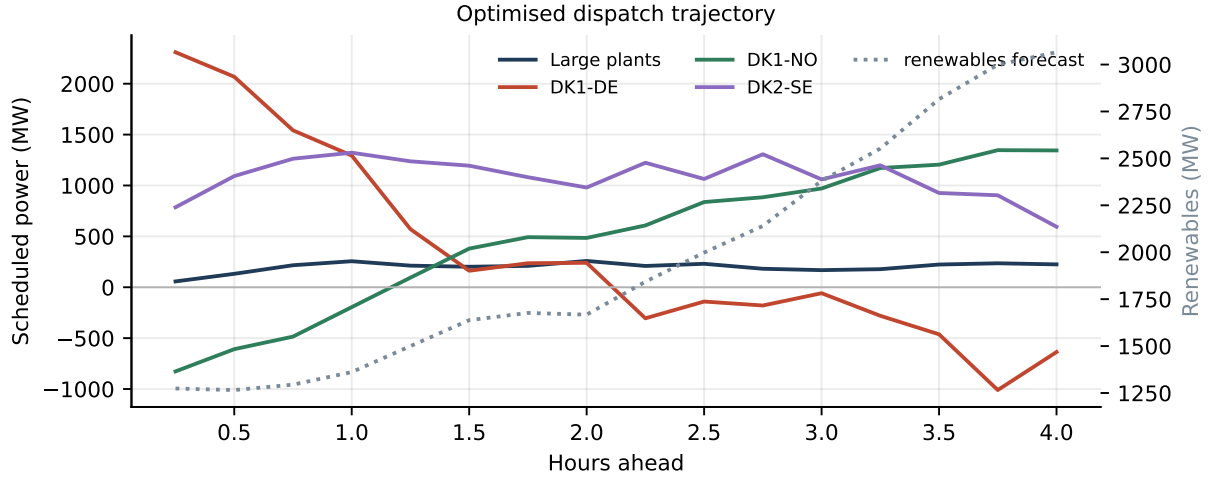


Figure 4: An optimized dispatch trajectory. As forecast wind rises through the afternoon the schedule withdraws from German imports, crossing into export, and shifts towards Norwegian hydro, while thermal output stays low. Every change respects its ramp limit.

same code now silently selects different columns. All feature selection here is by name. Separately, the deployed pipeline used a selection-pressure parameter of 150 where the notebook that validated the method used 8, which collapses the population to near-clones of a single individual within a few generations.

6 Limitations

The framework does not model unit commitment. Large-plant output is treated as a single aggregate quantity, where in reality it is dozens of units with individual minimum stable loads and start-up times.

Interconnector limits are fixed over the horizon. Real transfer capacities are published per market time unit and are frequently derated; a production system would consume them rather than assume them.

The model takes no account of exchange contract terms or energy prices, so a proposed schedule may be physically feasible and economically impossible.

The emissions model is compressive, as a regularized estimator fitted to a long-tailed target will be. Replaying the most recent 30 hours of observed states gives a mean absolute error of 9.9 g/kWh with a bias of -8.2 g/kWh: it over-predicts slightly at very low intensities and under-predicts at high ones. Absolute emission levels on the dashboard should therefore be read as estimates with roughly a 10 g/kWh error bar, not as measurements. The *reduction* is more robust than the levels, because the counterfactual and the optimized trajectory are evaluated with the same model and a shared bias largely divides out; but the ratio is itself compressed towards one, so the reported percentage is better treated as an indication of direction and rough magnitude than as a precise figure.

Most importantly, Energinet’s CO₂ figure is a *consumption-based* intensity for Denmark: it assigns emission factors to imports. A schedule can therefore lower the Danish number by importing more low-carbon power from Norway or Sweden without lowering emissions anywhere. Nordic hydro export capacity is finite, so power drawn to Denmark is power not delivered elsewhere, and the displaced consumer may burn gas instead. The reductions reported here are reductions in Denmark’s attributed intensity. Establishing how much of that is genuine abatement requires marginal emissions data for the surrounding system, which is outside the scope of this work. This

Issue	Correction
Optimizer exploited its own surrogate	Emissions model constrained monotone in every supply source; target changed from intensity to total emissions
Fully grown tree acted as a lookup table	Depth- and leaf-capped boosted ensemble; 25,863 leaves reduced to a bounded ensemble
No guard against unseen combinations	Leaf support and Mahalanobis novelty penalties in the fitness function
Shuffled splits on a time series	Chronological splits and time-series cross-validation throughout, with persistence baselines
Cosine similarity used for coherence	Explicit per-resource ramp limits derived from observed 15-minute changes
DK1-DK2 counted in the national balance	Restricted to the eight cross-border links, matching Energinet’s own convention
Results clamped so they could not look worse than the status quo	Removed; the counterfactual is published alongside the optimized result
Reduction measured against present emissions	Measured against the hold-current counterfactual over the same horizon
Independent per-step optimization	Trajectory chromosome with coupled ramp constraints
Search bounds fixed and stale	Bounds and ramp limits re-derived from the training year, capped by technical capacity

Table 2: Corrections applied to the original framework.

caveat applies equally to the original framework, where it was not stated.

7 Future work

Future work for customizing this model to any power grid might take into consideration other important aspects, like in-depth details about exchange contract terms, to the extent of their availability, and energy prices, and include them in the evaluation of solutions, extending the realism initiated by the ramp constraints.

Three directions follow from the limitations above. Replacing average with marginal emission factors would let the objective express abatement rather than attribution. Introducing semi-continuous constraints for minimum stable load would turn the problem into a genuine mixed-integer program, where the choice of an evolutionary heuristic could be benchmarked against an exact solver. And a systematic backtest, replaying historical periods and scoring proposed dispatch against realised outcomes, would convert the results reported here from a demonstration into evidence.

Visit the web app, which includes a dynamic comparison between recent emissions in Denmark’s power system, the hold-current counterfactual, and the minimized CO₂ levels over the next four hours, as well as the optimal energy resource distributions leading to these emission levels.